



Faculty of Computing and Technology
Department of Computing
Bachelor of Information Technology
End - Semester Examination
Year 2026, Semester 01 - Batch 03
SE 2105 – Advanced Databases (Practical)

Duration: 01 Hour

Instructions to the candidates:

1. Answer **ALL** the questions.
2. Create a folder using your Student ID and save all Notepad files inside the folder.
3. Upload the completed folder to the LMS.
4. The paper is marked out of 100 Marks.
5. Follow the General Guidelines given by the Department of Examination for exam Start time and closing time.

Question 01

- a) Write a query to create a database named EmployeeDB and select the newly created database for use.

(04 marks)

- b) Write a query to create a table named Employees with the following structure:

Attribute	Data Type
EmployeeID	INT
EmployeeName	VARCHAR(50)

Department	VARCHAR(30)
Salary	DECIMAL(10,2)

Set EmployeeID as the Primary Key.

(06 marks)

c) Write a query to insert the following records into the Employees table.

EmployeeID	EmployeeName	Department	Salary
101	Amal	IT	65000
102	Sara	Finance	55000
103	Nimal	IT	72000
104	David	HR	48000
105	Anu	Finance	60000

(05 marks)

d) Write SQL queries to:

i. Display all employees who belong to the IT department.

(03 Marks)

ii. Display employees whose salary is greater than 60,000.

(03 Marks)

iii. Display only the EmployeeName and Salary of all employees, ordered by salary from highest to lowest.

(04 marks)

[Total = 25 Marks]

Question 02

An online electronics store wants to manage its product information using MongoDB.

a) Write the command to create and select a MongoDB database named:

TechStoreDB

(03 Marks)

b) Write the command to create a collection named products.

(03 Marks)

c) Write the command to insert all of the following documents at once into the products collection.

productID: 101, productName: "Laptop", category: "Electronics", price: 250000, stock: 10 ,
productID: 102, productName: "Mouse", category: "Accessories", price: 3500, stock: 25 ,
productID: 103, productName: "Keyboard", category: "Accessories", price: 7500, stock: 15 ,
productID: 104, productName: "Monitor", category: "Electronics", price: 85000, stock: 8 ,
productID: 105, productName: "Speaker", category: "Electronics", price: 18000, stock: 20 ,
productID: 106, productName: "USB Cable", category: "Accessories", price: 1500, stock: 30

(10 marks)

d) Write the command to retrieve and display all documents from the products collection.

(04 marks)

e) Write the command to display the total number of documents available in the products collection

(05 marks)

[Total = 25 Marks]

Question 03

Using the products collection created in *Question 02*, write the commands to perform the following operations.

a) Retrieve all products where the price is greater than 10,000.

(04 Marks)

b) Retrieve all products where the category is "Electronics"

(03 Marks)

c) Retrieve products where:

- category is "Electronics"
- AND
- price is greater than 50,000

(05 Marks)

d) Display only the following fields for all products:

- productName
- price

Do not display the _id field.

(04 Marks)

e) Display all products sorted by price in descending order.

(04 Marks)

f) Display only the three most expensive products.

(05 marks)

[Total = 25 Marks]

Question 04

Continue using the products collection.

a) Write command to insert the following document into the collection:

```
{
  productID: 107,
  productName: "Smart Watch",
  category: "Electronics",
  price: 45000,
  stock: 12,
  supplier: {
    name: "TechWorld",
    country: "Japan"
  },
  reviews: [
    { user: "Kamal", rating: 5 },
    { user: "Sara", rating: 4 }
  ]
}
```

(05 marks)

b) Write the command to retrieve all products where the supplier's country is "Japan"

(04 Marks)

- c) Write the command to retrieve products that contain a review with a rating greater than 4.

(04 Marks)

- d) Write the command to increase the stock of all products by 5.

(04 Marks)

- e) Write the command to add the following new review to the reviews array of the Smart Watch:

`{ user: "Nimal", rating: 5 }`

(05 Marks)

- f) Write the command to delete the product where `productName = "USB Cable"`

(03 marks)

[Total = 25 Marks]

******* End of the Paper *******